

Use mathematical induction to show that,

$$1+2+\dots+n = \frac{n(n+1)}{2}$$

Proof

Basis step: ~~If n is a positive integer,~~

Let $p(1)$ is true. Then $\frac{1(1+1)}{2} = 1$

It shows that $p(1)$ is true.

Inductive step:

Let $p(k) = 1+2+\dots+k = \frac{k(k+1)}{2}$ — (1)
and $p(k)$ is true. i.e.,

$p(k) \rightarrow p(k+1)$ is true.

Therefore, $p(k+1) = 1+2+\dots+k+k+1 = \frac{(k+1)(k+2)}{2}$

L.H.S.

$$\begin{aligned} p(k+1) &= 1+2+\dots+k+k+1 \\ &= \frac{k(k+1)}{2} + (k+1) \quad [\text{from (1)}] \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+2)(k+1)}{2} = \text{R.H.S.} \end{aligned}$$

Hence, $P(k+1)$ is true. $\therefore$ not with 21
It proves the inductive step.

Thus, $1+2+\dots+n = \frac{n(n+1)}{2}$ [showed] 2A

Prove that, sum of the first n odd integers is n^2 .

Let, $P(n) = 1+3+5+\dots+(2n-1) = n^2$.

Basis step: ~~we~~ let $n=1$;

$P(1) = 1^2 = 1$ and it proves that the basis step is true.

Inductive step:

Let, $P(k) = 1+3+5+\dots+(2k-1) = k^2$ — ①

assume $P(k) \rightarrow P(k+1)$ is true.

$\therefore P(k+1) = 1+3+5+\dots+(2k-1)+(2k+1) = (k+1)^2$

L.H.S.

$1+3+5+\dots+(2k-1)+(2k+1)$

$= k^2 + 2k + 1$ [by ①]

$= (k+1)^2$

$= R.H.S.$

Hence $P(k+1)$ is true.

therefore $1+3+5+\dots+(2n-1) = n^2$ for n odd positive integers. [showed]

prove the inequality $n < 2^n$ for all positive integers n .

Let $P(n)$ be the proposition that $n < 2^n$.

Basis step:

$P(1)$ is true, because $1 < 2^1$. This completes the basis step.

Inductive step:

Let, $P(k)$ be the statement that ~~$k < 2^k$~~ $k < 2^k$. ①

assume, $P(k) \rightarrow P(k+1)$ is true.

$\therefore P(k+1)$ is the statement that $(k+1) < 2^{k+1}$.

Let's add 1 on the both side of (i), then $1 \leq 2^k$

$$\begin{aligned} \therefore k+1 &< 2^k + 1 \\ &\leq 2^k + 2^k \\ &\leq 2 \cdot 2^k \\ &\leq 2^{k+1} \end{aligned}$$

This shows that $(k+1)$ is true.

therefore, the inequality $n < 2^n$ for all positive integers n is true.

[proved]

Show that, $1 + 2 + 2^2 + 3^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n .

Let, $P(n) = 1 + 2 + 2^2 + 3^2 + \dots + 2^n = 2^{n+1} - 1$

Basis step:

$P(0) = 2^{0+1} - 1 = 1$ and it proves that the basis step is true.

Inductive step:

Let, $P(k) = 1 + 2 + 2^2 + \dots + 2^k \leftarrow 2^{k+1} - 1$ — (1)

Assume, $P(k) \rightarrow P(k+1)$ is true.

$$\therefore P(k+1) = 1 + 2 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{(k+1)+1} - 1$$

L.H.S.

$$1 + 2 + 2^2 + \dots + 2^k + 2^{k+1}$$

$$= 2^{k+1} - 1 + 2^{k+1}$$

$$= 2 \cdot 2^{k+1}$$

$$= 2^{k+2}$$

$\therefore$ R.H.S.

Hence, $P(k+1)$ is true.

Therefore, $1 + 2 + 2^2 + 3^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers of n . [Showed]